

```
(ht-set! numbers 1 "One")
(ht-set! numbers 2 "Two")
(ht-items (ht-select-keys numbers '(1)))

((1 "One"))
```

The following two examples are more comprehensive in using the hash table library functions. The *say-hello* function returns a greeting based on the name as shown below:

```
(defun say-hello (name)
  (let ((greetings (ht-create)))
    (ht-set! greetings "Adam" "Hello Adam!")
    (ht-set! greetings "Eve" "Hello Eve!")
    (ht-get greetings name "Hello stranger!")))

(say-hello "Adam")
"Hello Adam!"

(say-hello "Eve")
"Hello Eve!"

(say-hello "Bob")
"Hello stranger!"
```

The *ht* macro returns a hash table and we create nested hash tables in the following example:

```
(let ((alphabets (ht ("Greek" (ht (1 (ht ('letter "α")
                                      ('name "alpha")))
                                (2 (ht ('letter "β")
                                      ('name "beta")))))
  ("English" (ht (1 (ht ('letter "a")
                        ('name "A")))
                (2 (ht ('letter "b")
                        ('name "B"))))))))
  (ht-get* alphabets "Greek" 1 'letter))

"α"
```

Testing

The *ht.el* library has built-in tests that you can execute to validate the API functions. You first need to clone the repository using the following commands:

```
$ git clone git@github.com:Wilfred/ht.el.git
```

```
Cloning into 'ht.el'...
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
Receiving objects: 100% (471/471), 74.58 KiB | 658.00 KiB/s,
done.
remote: Total 471 (delta 0), reused 1 (delta 0), pack-reused 470
Resolving deltas: 100% (247/247), done.
```

If you do not have Cask, install the same using the instructions provided in the *README* file at <https://github.com/cask/cask>.

You can then change the directory into the cloned '*ht.el*' folder and run *cask install*. This will locally install the required dependencies for running the tests.

```
$ cd ht.el/
$ cask install
Loading package information... Select coding system (default
utf-8):
done
Package operations: 4 installs, 0 removals
- Installing [ 1/4] dash (2.12.0)... done
- Installing [ 2/4] ert-runner (latest)... done
- Installing [ 3/4] cl-lib (latest)... already present
- Installing [ 4/4] f (latest)... already present
```

A *Makefile* exists in the top-level directory and you can simply run '*make*' to run the tests, as shown below:

```
$ make
rm -f ht.elc
make unit
make[1]: Entering directory '/home/guest/ht.el'
cask exec ert-runner
.....

Ran 41 tests in 0.016 seconds
make[1]: Leaving directory '/home/guest/ht.el'
make compile
make[1]: Entering directory '/home/guest/ht.el'
cask exec emacs -Q -batch -f batch-byte-compile ht.el
make[1]: Leaving directory '/home/guest/ht.el'
make unit
make[1]: Entering directory '/home/guest/ht.el'
cask exec ert-runner
.....

Ran 41 tests in 0.015 seconds
make[1]: Leaving directory '/home/guest/ht.el'
make clean-elc
make[1]: Entering directory '/home/guest/ht.el'
rm -f ht.elc
make[1]: Leaving directory '/home/guest/ht.el'
```

You are encouraged to read the *ht.el* *README* file from the GitHub repository at <https://github.com/Wilfred/ht.el/blob/master/README.md> for more information. 

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